

Ex. No: 2

Recover Deleted or Damaged Files from a Storage Device Using TestDisk

Aim

To recover a missing partition and repair a corrupted partition on a storage device using the TestDisk data recovery tool.

Description

TestDisk is a free, open-source data recovery utility used to recover lost partitions and repair partition tables, boot sectors, and file systems that have become corrupted, damaged by software, or affected by certain viruses or human error. This exercise demonstrates the step-by-step procedure to recover a missing partition and repair a corrupted one using TestDisk.

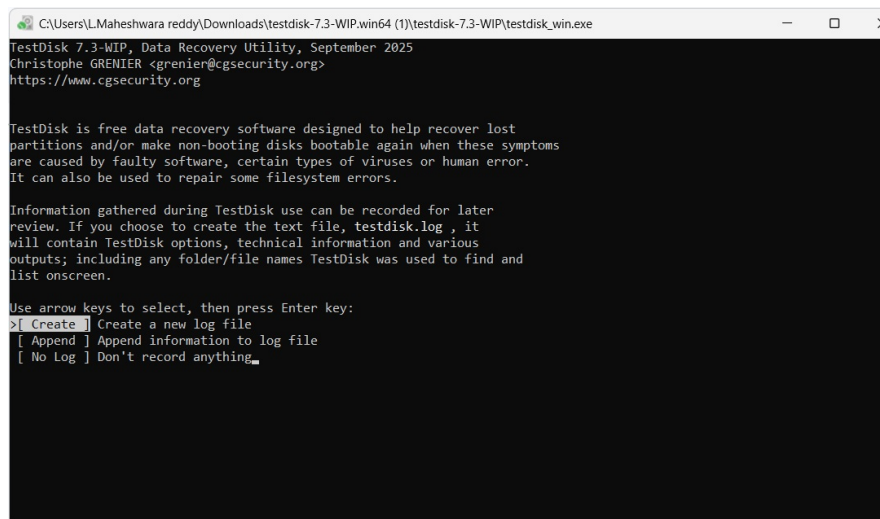
Procedure

Step 1: Log Creation

Choose Create to instruct TestDisk to create a log file containing technical information and messages, unless there is a reason to append data to an existing log or TestDisk is run from read-only media (in which case the log must be created elsewhere).

Choose None if messages and process details should not be written to a log file (useful when TestDisk is started from a read-only location).

Press Enter to proceed.



```
C:\Users\L.Maheshwara reddy\Downloads\testdisk-7.3-WIP.win64 (1)\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

TestDisk is free data recovery software designed to help recover lost
partitions and/or make non-booting disks bootable again when these symptoms
are caused by faulty software, certain types of viruses or human error.
It can also be used to repair some filesystem errors.

Information gathered during TestDisk use can be recorded for later
review. If you choose to create the text file, testdisk.log , it
will contain TestDisk options, technical information and various
outputs; including any folder/file names TestDisk was used to find and
list onscreen.

Use arrow keys to select, then press Enter key:
> [ Create ] Create a new log file
  [ Append ] Append information to log file
  [ No Log ] Don't record anything_
```

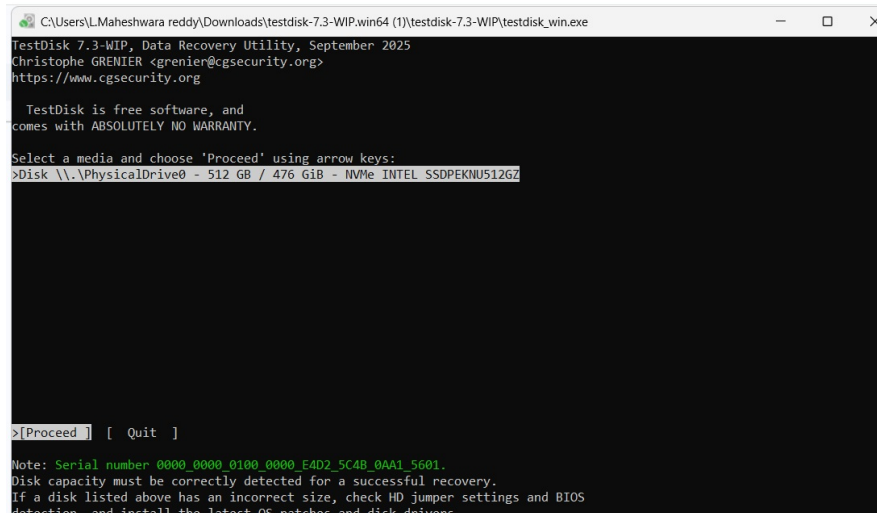
Step 2: Disk Selection

All hard drives are detected and listed with their correct size by TestDisk.

Use the Up/Down arrow keys to select the hard drive containing the lost partition(s).

Press Enter to proceed.

Note: If available, use the raw device `/dev/rdisk*` instead of `/dev/disk*` for faster data transfer.



```
CA\Users\L.Maheshwara reddy\Downloads\testdisk-7.3-WIP.win64 (1)\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

TestDisk is free software, and
comes with ABSOLUTELY NO WARRANTY.

Select a media and choose 'Proceed' using arrow keys:
>Disk \\.\PhysicalDrive0 - 512 GB / 476 GiB - NVMe INTEL SSDPEKN512GZ

>[Proceed] [ Quit ]

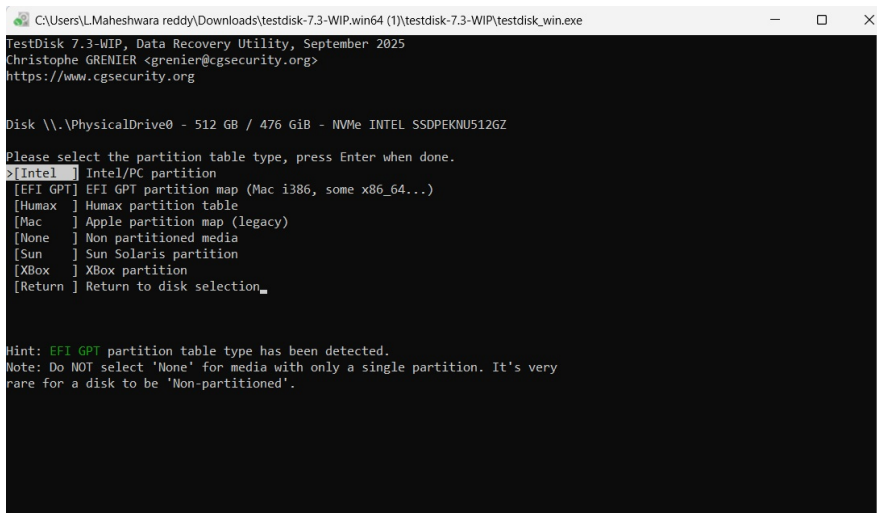
Note: Serial number 0000_0000_0100_0000_E4D2_5C4B_0AA1_5601.
Disk capacity must be correctly detected for a successful recovery.
If a disk listed above has an incorrect size, check HD jumper settings and BIOS
detection and install the latest OS patches and disk drivers.
```

Step 3: Partition Table Type Selection

TestDisk displays the available partition table types.

Select the partition table type — the default value auto-detected by TestDisk is usually correct.

Press Enter to proceed.



```
CA\Users\L.Maheshwara reddy\Downloads\testdisk-7.3-WIP.win64 (1)\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.\PhysicalDrive0 - 512 GB / 476 GiB - NVMe INTEL SSDPEKN512GZ

Please select the partition table type, press Enter when done.
>[Intel] Intel/PC partition
[EFI GPT] EFI GPT partition map (Mac i386, some x86_64...)
[Humax ] Humax partition table
[Mac ] Apple partition map (legacy)
[None ] Non partitioned media
[Sun ] Sun Solaris partition
[XBox ] Xbox partition
[Return] Return to disk selection_

Hint: EFI GPT partition table type has been detected.
Note: Do NOT select 'None' for media with only a single partition. It's very
rare for a disk to be 'Non-partitioned'.
```

Step 4: Current Partition Table Status

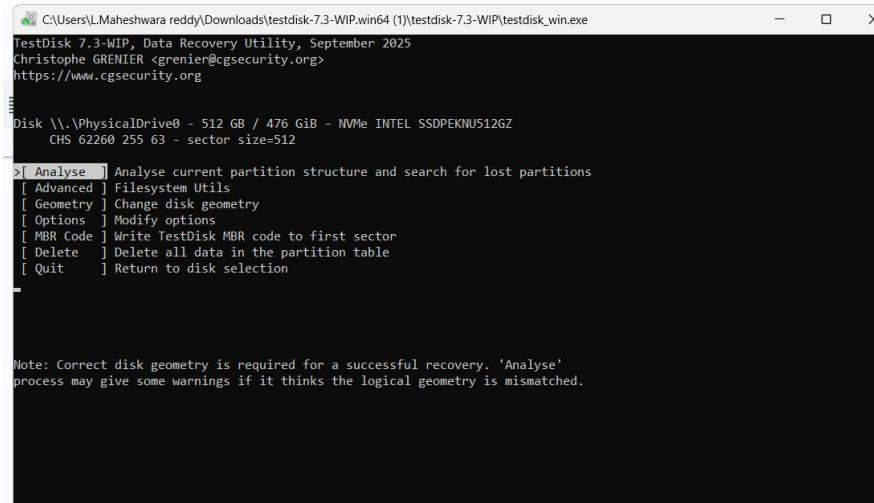
TestDisk displays its menu of options.

Use the default menu option Analyze to check the current partition structure and search for lost partitions.

Confirm Analyze by pressing Enter to proceed.

The current partition structure is now listed. Examine it for missing partitions and errors — for example, a partition listed twice indicates a corrupted partition or an invalid partition table entry, and an invalid NTFS boot message points to a faulty NTFS boot sector (a corrupted file system).

Confirm Quick Search to proceed.



```
C:\Users\L.Maheshwara reddy\Downloads\testdisk-7.3-WIP.win64 (1)\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.\PhysicalDrive0 - 512 GB / 476 GiB - NVMe INTEL SSDPEKNUS12GZ
CHS 62260 255 63 - sector size=512

> [Analyse]  Analyse current partition structure and search for lost partitions
[Advanced]  Filesystem Utils
[Geometry]  Change disk geometry
[Options]   Modify options
[MBR Code]  Write TestDisk MBR code to first sector
[Delete]    Delete all data in the partition table
[Quit]      Return to disk selection

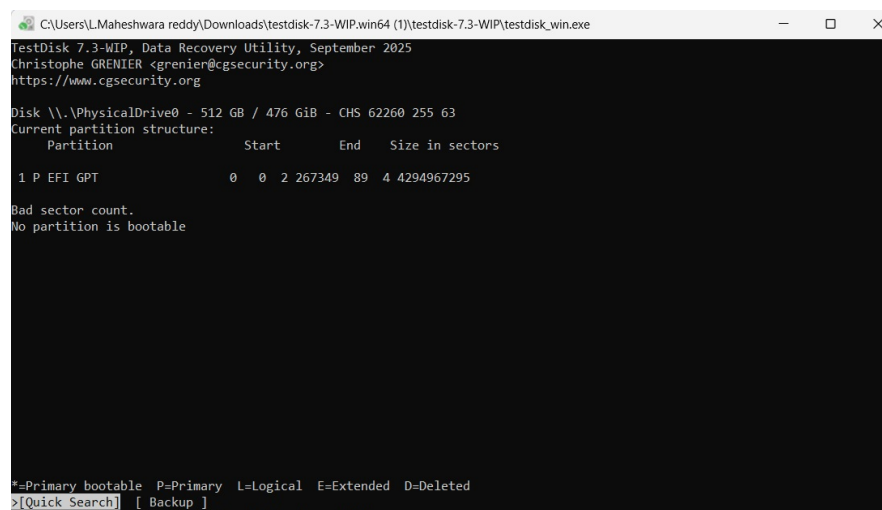
Note: Correct disk geometry is required for a successful recovery. 'Analyse'
process may give some warnings if it thinks the logical geometry is mismatched.
```

Step 5: Quick Search for Partitions

TestDisk displays results in real time as it performs the Quick Search.

Once a missing partition is found, highlight it and press ‘p’ to list its files (press ‘q’ to quit back to the previous display). Deleted entries are shown in red.

Verify that all directories and data are correctly listed, then press Enter to proceed.



```
C:\Users\L.Maheshwara reddy\Downloads\testdisk-7.3-WIP.win64 (1)\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.\PhysicalDrive0 - 512 GB / 476 GiB - CHS 62260 255 63
Current partition structure:
  Partition      Start      End      Size in sectors
  1 P EFI GPT      0  0  2 267349  89  4 4294967295

Bad sector count.
No partition is bootable

*=Primary bootable P=Primary L=Logical E=Extended D=Deleted
>[Quick Search] [Backup]
```

Step 6: Save Partition Table or Search for More Partitions

If all partitions are available and the data is correctly listed, go to the Write menu to save the partition structure. The Extd Part menu allows choosing whether the extended partition uses all available disk space or only the minimal required space.

If a partition is still missing, highlight Deeper Search and press Enter to proceed.

```
C:\Users\L.Maheshwara reddy\Downloads\testdisk-7.3-WIP.win64 (1)\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.\PhysicalDrive0 - 512 GB / 476 GiB - CHS 62260 255 63
Partition      Start      End      Size in sectors
* FAT32        0 32 33    33 69 36    532480 [EFI System Partition] [SYSTEM]
P HPFS - NTFS   62088 95 24 62227 84 1    2232320
P FAT32 LBA     62227 84 2 62260 121 5    532480 [MYASUS]

Structure: Ok. Use Up/Down Arrow keys to select partition.
Use Left/Right Arrow keys to CHANGE partition characteristics:
*=Primary bootable P=Primary L=Logical E=Extended D=Deleted
Keys A: add partition, L: load backup, T: change type, P: list files,
Enter: to continue
FAT32, blocksize=4096, 272 MB / 260 MiB
```

Step 7: Deeper Search (for a Still-Missing Partition)

Deeper Search also looks for the FAT32 backup boot sector, NTFS backup boot superblock, and ext2/ext3 backup superblock to detect more partitions by scanning each cylinder.

After the Deeper Search, review the results — for example, a partition found using its backup boot sector, or a partition shown twice with different sizes because the two entries overlap.

Partitions listed with status D (Deleted) will not be recovered unless their status is changed. When two entries overlap, identify which one is the correct partition to recover.

Highlight each candidate partition and press ‘p’ to list its files to check whether the file system is damaged or intact; press ‘q’ to return to the previous display.

Leave the partition with a damaged file system marked as D (Deleted), and confirm that the correct partition (with its files listed properly) is the one to recover.

```
C:\Users\L.Maheshwara reddy\Downloads\testdisk-7.3-WIP.win64 (1)\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.\PhysicalDrive0 - 512 GB / 476 GiB - CHS 62260 255 63
Partition      Start      End      Size in sectors
1 * FAT32        0 32 33    33 69 36    532480 [EFI System Partition] [SYSTEM]
2 P HPFS - NTFS   62088 95 24 62227 84 1    2232320
3 P FAT32 LBA     62227 84 2 62260 121 5    532480 [MYASUS]

[ Quit ] [ Return ] >[Deeper Search] [ Write ]
Try to find more partitions
```

Step 8: Change Partition Status

The available statuses are Primary, * (bootable), Logical, and Deleted.

Using the Left/Right arrow keys, change the status of the correct partition from D (Deleted) to L (Logical) so that it can be recovered.

Note: If a partition is listed as *(bootable) but is not the boot partition, its status can be changed to Primary.

Press Enter to proceed.

```
C:\Users\L.Maheshwara reddy\Downloads\testdisk-7.3-WIP.win64 (1)\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.\PhysicalDrive0 - 512 GB / 476 GiB - CHS 62260 255 63
Partition      Start      End      Size in sectors
* FAT32        0 32 33    33 69 36    532480 [EFI System Partition] [SYSTEM]
P HPFS - NTFS   62088 95 24 62227 84 1    2232320
P FAT32 LBA     62227 84 2 62260 121 5    532480 [MYASUS]

Structure: Ok. Use Up/Down Arrow keys to select partition.
Use Left/Right Arrow keys to CHANGE partition characteristics:
* = Primary bootable P = Primary L = Logical E = Extended D = Deleted
Keys A: add partition, L: load backup, T: change type, P: list files,
Enter: to continue
FAT32, blocksize=4096, 272 MB / 260 MiB
```

Step 9: Partition Table Recovery

It is now possible to write the new partition structure. The extended partition is set automatically, as TestDisk recognises it from the differing partition structure.

Once all partitions are listed correctly, confirm by selecting Write, then press Enter, 'y' to confirm, and OK.

The partitions are now registered in the partition table.

```
C:\Users\L.Maheshwara reddy\Downloads\testdisk-7.3-WIP.win64 (1)\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
P HPFS - NTFS   62088 95 24 62227 84 1    2232320
Directory /
>dr-xr-xr-x    0 0      0 28-Jul-2026 10:33 .
dr-xr-xr-x    0 0      0 28-Jul-2026 10:33 ..
dr-xr-xr-x    0 0      0 28-Jul-2026 10:41 Recovery
dr-xr-xr-x    0 0      0 17-Feb-2025 18:16 System Volume Information
-r--r--r--    0 0      0 29-Jul-2026 12:12 $WINRE_BACKUP_PARTITION.MARKER

Next
Use Right to change directory, 'h' to hide Alternate Data Stream
'q' to quit, ':' to select the current file, 'a' to select all files
'c' to copy the selected files 'C' to copy the current file
```

Step 10: NTFS Boot Sector Recovery

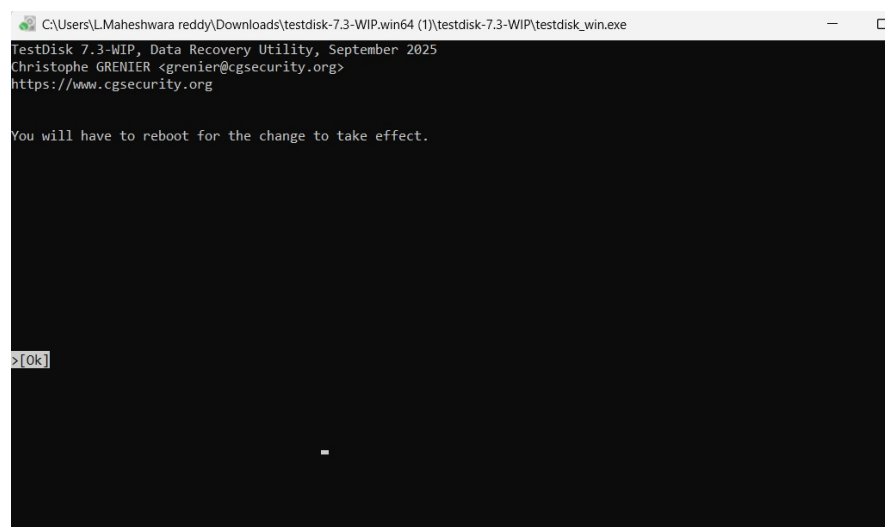
If the boot sector of a partition (e.g., Partition 1) is still damaged, it must be repaired. When the boot sector status is Bad and the backup boot sector is Valid, the two boot sectors are not identical.

Select Backup BS to copy the backup boot sector over the damaged boot sector, validate with Enter, confirm with 'y', and select OK.

A confirmation message indicates that the boot sector and its backup are now identical — the NTFS boot sector has been successfully recovered.

Press Enter to quit.

TestDisk will prompt that the computer must be restarted to access the recovered data. Press Enter one last time and reboot the computer.



Result

The missing partition was successfully identified and recovered, the partition table was rewritten with the correct partition structure, and the damaged NTFS boot sector was repaired using its backup boot sector. Thus, the lost data was successfully recovered using TestDisk.